

PPOM1 Lecture # 71a: OMM: Basics of Diagnosis: Acromioclavicular (AC), Sternoclavicular (SC), and Glenohumeral (GH) Joints

Week 8, Feb 2026

**To Shan Li, DO
Professor
tli05@nyit.edu**

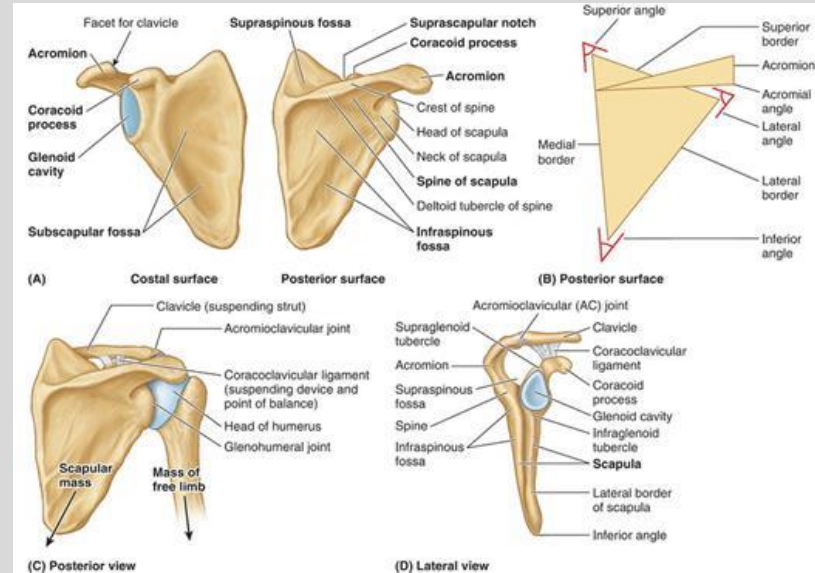
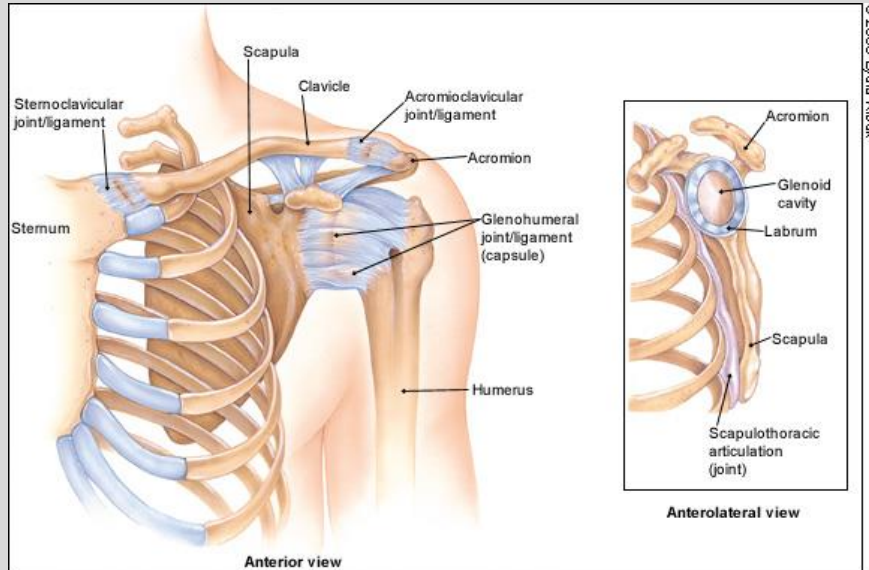
Session Objectives

1. diagnose somatic dysfunction of the AC joint
2. diagnose somatic dysfunction of the SC joint
3. diagnose somatic dysfunction of the GH joint

The AC, SC, and GH joints

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Woodward TW & Best TM, American Family Physician, 2006;
61(10)

Department of Osteopathic Manipulative Medicine

Moore et al, Clinically Oriented Anatomy 8e, Chp. 3: Upper Limb
<https://login.nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=168701099&bookid=2212>

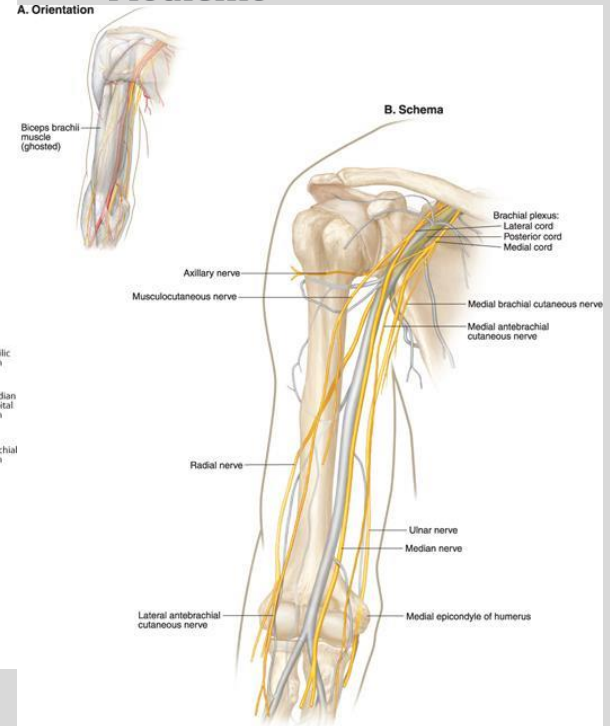
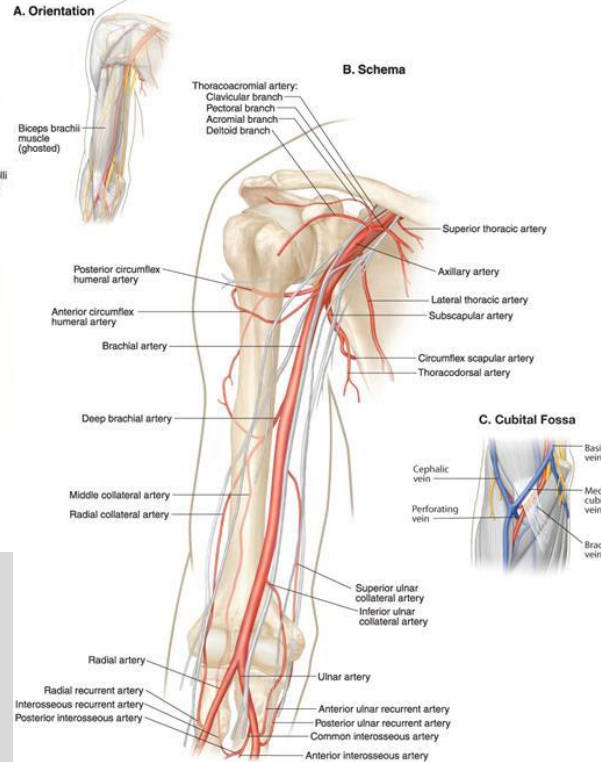
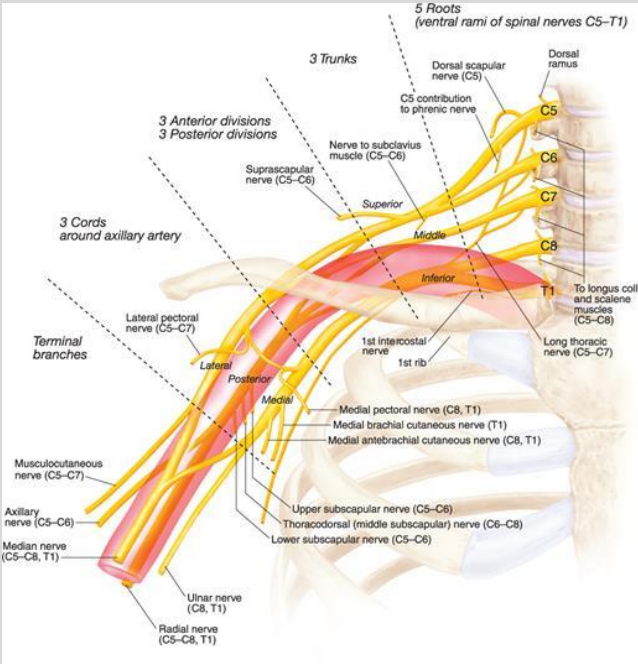
NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine



Gest T, Lippincott Atlas of Anatomy 2e, Chp. 2: The Upper Limb,

<https://login.nyit.idm.oclc.org/login?url=https://premiumbasicsscience.lwwhealthlibrary.com/content.aspx?sectionid=249368698&bookid=2991>

Glides of Clavicle at SC joint

- Shrug shoulders = glides inferiorly = abduction SD



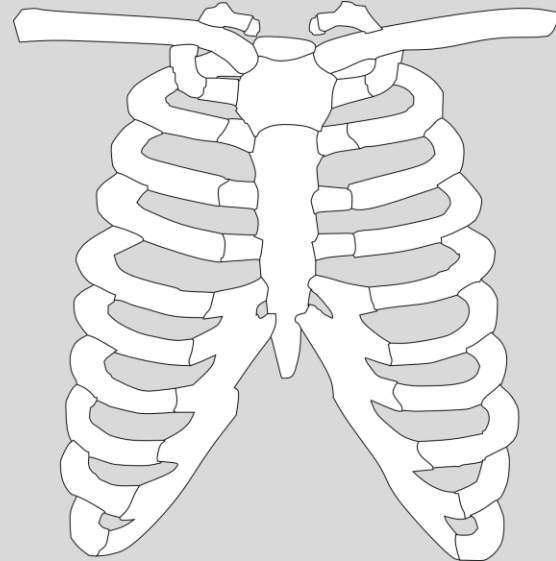
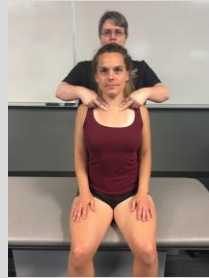
Protract shoulders = posterior glide = flexion SD



- Depress shoulders = glides superiorly = adduction SD



- Retract shoulders = anterior glide = extension SD



Summary on SC joint glides and SD

- Reciprocal motion is present at the clavicles with shoulder depression inducing superior glide at the SC joints, shoulder elevation inducing inferior glide at the SC, shoulder protraction inducing posterior glide at the SC, and shoulder retraction inducing anterior glide at the SC joints.
- Somatic dysfunction at the SC is often a combination of anterior or posterior and superior or inferior somatic dysfunctions.

Diagnose via assessing glide motion at the SC joints during active shoulder motion or via passive motion testing at SC joint motion:

Diagnosis	Active motion	Passive motion
Anterior/extension somatic dysfunction	Glides anterior easily with backward shrug (shoulder retraction), resists posterior glide with forward shrug (shoulder protraction)	Marked resistance to anterior to posterior pressure on proximal clavicle
Posterior/flexion SD	Glides posterior easily with forwards shrug, resists anterior glide with backward shrug	Glides easily with anterior to posterior pressure on proximal clavicle
Superior/adduction SD	Glides superior easily with downward shrug (shoulder depression), resists inferior glide with upward shrug (shoulder elevation)	Marked resistance to superior to inferior pressure on proximal clavicle
Inferior/abduction SD	Glides inferior easily with upward shrug, resists superior glide with downward shrug	Glides easily with superior to inferior pressure on proximal clavicle

Naming AC joint diagnosis

- **Abduction** somatic dysfunction (ie. adduction restriction)
- **Adduction** somatic dysfunction (ie. abduction restriction)
- **Internal rotation** somatic dysfunction (ie. external rotation restriction)
- **External rotation** somatic dysfunction (ie. internal rotation restriction)
- **Elevated** somatic dysfunction = superior glide preference = step down
- **Depressed** somatic dysfunction = inferior glide preference = step up



The 3 glides go together:

- Abduction, Internal rotation, and Superior glide (ie. superior glide of clavicle relative to acromion)
- Adduction, External rotation, and Inferior glide (ie. inferior glide of clavicle relative to acromion)

- Be sure to compare both sides (remember that it is also possible to have dysfunction on both sides, so include other criteria of TART to determine if there is somatic dysfunction)

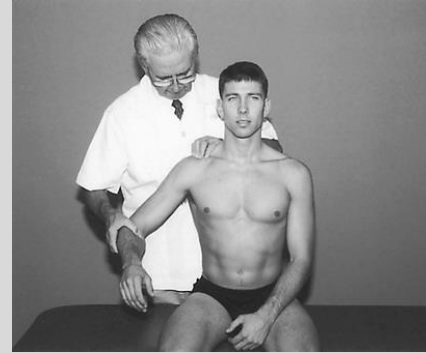
Diagnose via passive motion testing at the AC joint:

Diagnosis	Motion	Position
Inferior Glide (Step UP) Depressed SD	Clavicle springs easily	Clavicle inferior to the acromion
Superior Glide (Step DOWN) Elevated SD	Resists springing	Clavicle superior to the acromion

Diagnosis of AC joint- Abduction and Adduction



Abduction & adduction testing



Abduction, superior glide, and internal rotation testing

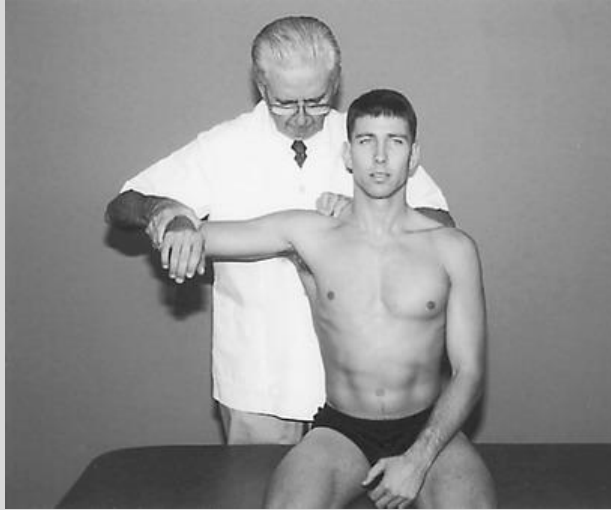


Adduction, inferior glide, and external rotation testing

Greenman's Principles of Manual Medicine, 5th edition, Ch. 18

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=116100818&bookid=1743#116100895>

Diagnosis of AC joint- Internal and External rotation



Externally and Internally rotate the shoulder from this position
while monitoring the AC joint

*you can feel the clavicle glide superior with internal rotation and inferior with external rotation with this test

Greenman's Principles of Manual Medicine, 5th edition, Ch. 18

<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=116100818&bookid=1743#116100895>

Dx of GH joint

Diagnose via passive gross range of motion testing (flexion, extension, internal rotation, external rotation, abduction, and adduction) or via passive motion testing of the minor glides at the humeral head:



Dx GH joint continued

NEW YORK INSTITUTE

Diagnosis	Gross ROM	Minor glides
Anterior/ extension SD	As the shoulder extends, horizontal abducts, or externally rotates, the humeral head moves easily anteriorly	Passively gliding humerus anteriorly meets no resistance until anatomic barrier is reached
Posterior/ flexion SD	As the shoulder flexes, horizontally adducts, or internally rotates, the humeral head moves easily posteriorly	Passively gliding humerus posteriorly meets no resistance until anatomic barrier is reached
Superior/ adduction SD	As the shoulder elevates (shrugs up) or adducts, the humeral head moves easily superiorly	Passively gliding humerus superiorly meets no resistance until anatomic barrier is reached
Inferior/abduction SD	As the shoulder depresses (pulled down inferiorly) or abducts, the humeral head moves easily inferiorly	Passively gliding humerus inferiorly meets no resistance until anatomic barrier is reached

Final points

1. Proper functioning of the AC, SC, and GH joints all contribute to proper functioning of the shoulder
2. Abnormalities of motion of these joints can generate pain, restrict motion, and potentially contribute to impingement of other structures (eg. tendons, nerves, and vasculature)
3. Motion of these joints is normal. It is abnormal motion that we are naming, using the convention of naming the freedom.

- There will be a worksheet posted to help guide you gain competency with the material. Also posted will be the answer sheet. You do not have to turn in.

References

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

1. Gest, Thomas, Lippincott Atlas of Anatomy, 2e, Wolters Kluwer, 2020: Chp. 2, The Upper Limb
2. Moore Keith L, Dalley Arthur F, Agur Anne MR, Clinically Oriented Anatomy, 8e, Wolters Kluwer, 2018, Chapter 3: Upper Limb
3. Pixabay images
4. Woodward Thomas W, Best Thomas M. The Painful Shoulder: Part I. Clinical Evaluation, American Family Physician. 2000;61 (10):3079-3088

Thank You

Lecture and Lab Feedback Form:

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>